==
ROOT CAUSE ANALYSIS FRAMEWORK
ADCS TELEMETRY FAULT DIAGNOSIS ? KNOWLEDGE BASE
ESA OPS-SAT Mission Operations
Version 1.0
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1. INTRODUCTION TO ROOT CAUSE ANALYSIS FOR SATELLITE ADCS

Root Cause Analysis (RCA) in satellite ADCS operations is the process of determining the underlying physical, electrical, or software cause of an observed telemetry anomaly. Unlike threshold-based alarming (which only detects that something is wrong), RCA explains WHY it is wrong and WHAT the operator should do about it.

The ADCS RCA framework for OPS-SAT uses the following logic chain:

  OBSERVED SYMPTOM (telemetry channels with anomalies)
       ?
  ANOMALY PATTERN (type, affected channels, statistical signature)
       ?
  CAUSAL HYPOTHESIS (physical mechanism that produces this pattern)
       ?
  CONFIRMING EVIDENCE (cross-subsystem telemetry, orbital context)
       ?
  ROOT CAUSE ASSIGNMENT (specific hardware/software/environment cause)
       ?
  CORRECTIVE ACTION (targeted intervention to restore nominal state)

2. CAUSAL DEPENDENCY MAP

The following dependency relationships determine how faults propagate through the ADCS subsystem chain:

MAGNETOMETER CHANNELS (CADC0872?CADC0874)
  Affected by (upstream causes):
  ? Power subsystem (EPS): Voltage fluctuations, switching noise
  ? Magnetorquers: EMI from active torquer coils
  ? Radiation environment: SEU, TID accumulation
  ? OBC software: Calibration table errors, I2C driver faults
  ? Physical sensor: Bias drift, gain change, physical damage

  Affects (downstream consequences):
  ? Attitude determination accuracy
  ? Magnetorquer control effectiveness
  ? Kalman filter state estimate quality
  ? Pointing accuracy for all subsystems

PHOTODIODE CHANNELS (CADC0875?CADC0880)
  Affected by (upstream causes):
  ? Orbital geometry: Eclipse, sun angle, Earth albedo
  ? Attitude state: Current satellite orientation
  ? Radiation environment: Photodiode junction degradation, window darkening
  ? Physical solar arrays/panels: Shadows, unexpected deployments
  ? ADC electronics: Signal chain faults

  Affects (downstream consequences):
  ? Sun vector estimation accuracy

  ? Solar power generation attitude optimisation
  ? Safe mode sun-pointing capability
  ? Power budget during power-critical operations

REACTION WHEELS (affects all channels indirectly)
  If reaction wheels are saturated or faulty:
  ? Satellite may begin drifting from target attitude
  ? Over time, this causes attitude-dependent variation in all channels
  ? Magnetometer readings drift as satellite moves through orbital field
  ? Photodiode readings change as satellite points in different direction

3. ANOMALY SIGNATURE LIBRARY
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Each anomaly pattern has a characteristic signature in the 18 statistical
features used by the FCNN model. This library maps feature profiles to
root causes.

SIGNATURE 1: SENSOR BIAS DRIFT (Magnetometer)
  Feature Profile:
    - mean (?): ELEVATED or DEPRESSED relative to nominal
    - variance (?²): NORMAL (bias does not increase variance)
    - skewness (??): NEAR ZERO (symmetric distribution, just shifted)
    - kurtosis (??): NEAR ZERO (normal distribution shape)
    - slope: SMALL (bias is approximately constant)
    - autocorrelation (??): HIGH (>0.95, signal is still smooth)
    - entropy: NORMAL (pattern complexity unchanged)
  Diagnostic Confidence: HIGH
  Channels Typically Affected: 1 (single-axis bias) or all 3 (systematic)

SIGNATURE 2: SENSOR STUCK VALUE (Frozen Reading)
  Feature Profile:
    - variance (?²): NEAR ZERO (< 1% of normal)
    - range (x_max - x_min): NEAR ZERO
    - zero-crossing rate: ZERO
    - autocorrelation (??): ~1.0
    - entropy: MINIMUM (fully predictable)
    - slope: ~0.0
  Diagnostic Confidence: VERY HIGH
  Channels Typically Affected: 1 (single-channel ADC fault)

SIGNATURE 3: HIGH NOISE / EMI INTERFERENCE
  Feature Profile:
    - variance (?²): HIGHLY ELEVATED (5-20× normal)
    - mean absolute deviation (MAD): ELEVATED
    - first-difference mean (??): ELEVATED
    - zero-crossing rate: ELEVATED
    - autocorrelation (??): LOW (< 0.3 for heavy noise)
    - entropy: ELEVATED
  Diagnostic Confidence: HIGH
  Channels Typically Affected: 1-3, correlated with EMI source

SIGNATURE 4: SATELLITE TUMBLING (All channels affected)
  Feature Profile (MAGNETOMETER):
    - variance (?²): VERY HIGH
    - range: VERY LARGE (full ±60 ?T range being swept through)
    - zero-crossing rate: HIGH (oscillating through positive and negative)
    - entropy: HIGH
    - signal appears sinusoidal or complex periodic
  Feature Profile (PHOTODIODE):

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    - variance: HIGH (flashing between light and dark)
    - zero-crossing rate: HIGH
    - multiple channels cycling on and off in synchrony
  Diagnostic Confidence: VERY HIGH when seen in both sensor types
  Channels Typically Affected: ALL channels simultaneously

SIGNATURE 5: ECLIPSE TRANSITION (Normal condition ? do not alarm)
  Feature Profile (PHOTODIODE):
    - Rapid decrease in mean from sunlit level to near zero
    - Occurs on all channels within 5-10 seconds of transition
    - Magnetometer channels: NO significant change (field is continuous)
  Diagnostic Confidence: HIGH if correlated with orbital prediction
  Channels: CADC0875?CADC0880 all drop to zero simultaneously

SIGNATURE 6: REACTION WHEEL SATURATION
  Feature Profile:
    - Gradual change in magnetometer readings as attitude drifts slowly
    - No high-frequency noise (not EMI-related)
    - Slope (linear trend) visible in magnetometer channels
    - Photodiode channels change gradually consistent with attitude drift
  Diagnostic Confidence: MEDIUM (requires cross-correlation with RW TM)
  Channels: Gradual drift in magnetometer and selected photodiodes

SIGNATURE 7: PHOTODIODE ELECTRICAL FAILURE
  Feature Profile:
    - Single channel near zero (not consistent with current attitude)
    - Other channels show expected sunlit values
    - No orbital explanation (not eclipse, not pointing away from Sun)
    - Variance near zero (not flickering ? firmly stuck at zero)
  Diagnostic Confidence: HIGH after ruling out orbital/geometric cause
  Channels: Single CADC channel

SIGNATURE 8: POWER SUPPLY ANOMALY AFFECTING SENSORS
  Feature Profile:
    - Multiple channels simultaneously showing degraded readings
    - May present as elevated noise on ALL analog sensor channels
    - Correlated with EPS voltage anomaly in housekeeping telemetry
    - May include transient spikes or dropouts
  Diagnostic Confidence: MEDIUM (requires EPS telemetry cross-check)
  Channels: Multiple channels across both magnetometers and photodiodes

4. CROSS-CHANNEL CORRELATION RULES
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The pattern of which channels are affected together provides strong
diagnostic information:

RULE 1: Only CADC0872 (Mag X) affected ? Single-axis sensor or EMI issue
RULE 2: CADC0872 + CADC0873 + CADC0874 all affected ? Common cause
         (power supply, EMI source, radiation event, tumbling)
RULE 3: Only CADC0875?CADC0880 affected ? Solar/attitude/EPS issue
RULE 4: Both magnetometers AND photodiodes affected ? Tumbling, power event,
         or major spacecraft event
RULE 5: Magnetometers noisy + Photodiodes normal ? EMI from torquers or EPS
RULE 6: Magnetometers normal + Photodiodes anomalous ? Sun geometry issue
         or individual photodiode failure

5. CONFIDENCE SCORING FOR ROOT CAUSE HYPOTHESES
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When assigning confidence to a root cause hypothesis, use the following
scoring matrix:

EVIDENCE WEIGHT:
  - Feature signature matches perfectly: +3 points
  - Feature signature partially matches: +2 points
  - Temporal correlation with known event (torquer, EPS, eclipse): +3 points
  - Cross-channel correlation pattern matches: +2 points
  - Orbital/environmental context matches: +2 points
  - Historical precedent (same anomaly before): +2 points
  - Contradicting evidence: -3 points per contradiction

CONFIDENCE THRESHOLD:
  Score 8+: HIGH confidence ? proceed with corrective action
  Score 5-7: MEDIUM confidence ? proceed with caution, monitor result
  Score 3-4: LOW confidence ? perform additional diagnostic checks first
  Score < 3: INSUFFICIENT evidence ? escalate to engineering team

6. ENVIRONMENTAL CONTEXT FACTORS
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The following external conditions systematically affect ADCS telemetry
and must be accounted for in root cause analysis:

SOUTH ATLANTIC ANOMALY (SAA):
  Region: Approximately -50° to +10° latitude, -90° to +30° longitude
  Effect: Elevated trapped proton/electron flux. SEU rates increase 10-100×.
  ADCS impact: Transient spikes in all ADC channels. Single-sample outliers
  are NORMAL during SAA passes. Multi-sample anomalies during SAA still
  require investigation.
  Duration: Approximately 10-15 minutes per SAA pass.

GEOMAGNETIC STORMS (Kp > 4):
  Effect: Earth's magnetic field intensity and direction varies rapidly.
  ADCS impact: Magnetometer readings vary more than predicted by IGRF model.
  False alarms may be generated by anomaly detection systems. Check Kp index
  (available from NOAA Space Weather Prediction Center) before diagnosing
  magnetometer anomalies.

ECLIPSE/TERMINATOR:
  Effect: Rapid thermal cycling and loss of solar power.
  ADCS impact: Temperature changes may cause thermal expansion/contraction
  affecting physical sensor alignment. Magnetometer bias may shift slightly.
  EPS mode changes at eclipse entry/exit may cause transient EMI.

ORBITAL MANOEUVRES (if any):
  Effect: If propulsion is available, thruster firings create vibration and
  electromagnetic transients.
  ADCS impact: All sensor channels may show transient anomalies during
  firing events. Normal. Clear within 30 seconds of firing completion.

7. QUICK REFERENCE: CHANNEL-TO-CAUSE MAPPING
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Anomaly Channel Pattern	Most Likely Root Cause	Priority Action
CADC0872 bias shift only	X-axis EMI / calibration	CA-M01
CADC0873 bias shift only	Y-axis EMI / calibration	CA-M01
CADC0874 bias shift only	Z-axis EMI / calibration	CA-M01
All three mag bias shifted	Common cause: EPS/radiation	CA-M01 + EPS check

One mag stuck at constant	ADC fault / I2C error	CA-M02 ? CA-M03
All mags high noise	Magnetorquer EMI or tumbling	Check torquer status
CADC0875 dark (not eclipse)	Photodiode failure or shadow	CA-P01
All PDs dark (not eclipse)	Tumbling / total power failure	CA-S01
All PDs noisy simultaneously	Tumbling or EPS anomaly	Check ADCS rates
Mixed mags + PD anomalies	Tumbling / major power event	CA-S01 (emergency)

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END OF ROOT CAUSE ANALYSIS FRAMEWORK
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